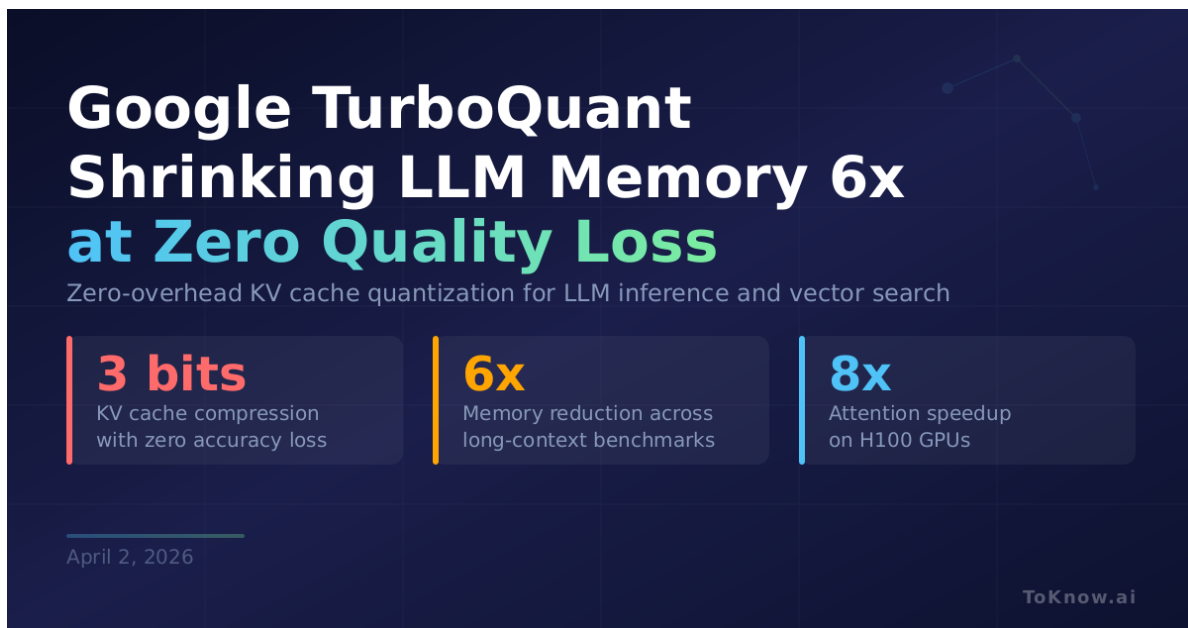


Google TurboQuant: Shrinking LLM Memory 6x at 3 Bits With Zero Quality Loss

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The graphic features a dark blue background with a subtle grid pattern. In the top right corner, there is a faint line graph with three data points connected by lines. The main title 'Google TurboQuant Shrinking LLM Memory 6x at Zero Quality Loss' is prominently displayed in white and green text. Below the title, a subtitle reads 'Zero-overhead KV cache quantization for LLM inference and vector search'. Three key performance indicators are highlighted in separate boxes: '3 bits' for KV cache compression, '6x' for memory reduction, and '8x' for attention speedup. The date 'April 2, 2026' is shown in the bottom left, and the 'ToKnow.ai' logo is in the bottom right.

Google TurboQuant Shrinking LLM Memory 6x at Zero Quality Loss

Zero-overhead KV cache quantization for LLM inference and vector search

- 3 bits**
KV cache compression
with zero accuracy loss
- 6x**
Memory reduction across
long-context benchmarks
- 8x**
Attention speedup
on H100 GPUs

April 2, 2026

ToKnow.ai

Google Research introduced [TurboQuant](#), a compression algorithm that shrinks the KV cache (the working memory LLMs use to store context during generation) to just 3 bits per value with zero accuracy loss and no retraining. It combines two techniques. [PolarQuant](#) converts key-value vectors from standard coordinates to polar coordinates, exploiting the predictable distribution of angles to skip the normalization step that normally adds 1-2 extra bits of

overhead. [QJL](#) then applies a 1-bit mathematical transform to the leftover error, correcting estimation bias at near-zero cost. Tested on Gemma and Mistral across [LongBench](#), Needle-in-Haystack, [RULER](#), and ZeroSCROLLS, TurboQuant achieves over 6x memory reduction while maintaining perfect scores on needle-in-haystack retrieval tasks. On H100 GPUs, 4-bit TurboQuant computes attention up to 8x faster than unquantized baselines. All three papers include theoretical proofs showing the methods operate within a 2.7x factor of information-theoretic lower bounds.

The KV cache is the main memory bottleneck when serving LLMs, especially with [long context windows](#). A 6x reduction means a server handling 10 concurrent users can now handle 60, or a model that needed multiple GPUs fits on one. TurboQuant is data-oblivious and needs no fine-tuning, so it applies to any LLM deployment out of the box. The same techniques also speed up vector search index building, making large-scale semantic search cheaper.

The pattern here matches other recent efficiency work: mathematical structure, not bigger hardware, is unlocking the next round of deployment gains. Between [Nemotron-Cascade 2's](#) 20x parameter reduction through mixture-of-experts and TurboQuant's 6x memory compression, the cost of running frontier-level AI is dropping faster through algorithms than through chip improvements.

Sources: - [Google Research Blog: TurboQuant](#) - [TurboQuant Paper \(ICLR 2026\)](#) - [PolarQuant Paper \(AISTATS 2026\)](#) - [QJL Paper \(AAAI\)](#)

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